

Final term Question paper

— 2024 —

"Objective part":

① if $T: X \rightarrow Y$ is an Isometric Isomorphism
b/w two norm space X and Y , then $\Rightarrow$.

T is bijective and norm preserving.

② The dual space of a normed space X contain
all — in X .

Bounded Linear function

③

all — in X .

Bounded Linear function

③ Every inner product space — space.

Metric, Normed, Topological, All

④ Every complete inner product space is — space.

Hilbert

⑧ Every complete normed space is — space.

Banach.

⑨ If inner product space $X_n \rightarrow X$ and $Y_n \rightarrow Y$, then —

(9) If inner product space $X_n \rightarrow X$ and $Y_n \rightarrow Y$, then —

$$\langle x_n, y_n \rangle = \langle x, y \rangle$$

(10) An orthonormal set is linearly —
Dependent

(11) The Pythagorean relation is. —

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

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MATHS

(13) Inner Product is the generalization —

Dot Product

(14) Any Pair of vector in an orthogonal system is —

(15) The dual space of ℓ^1 is: —
 ℓ^∞

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 ℓ^∞

(16) The space — is a hilbert space.

Inner Product

(17) The space $C[a, b]$ is: —

collection of all continuous function define on $[a, b]$.

(18) $\|x+y\| \leq \|x\| + \|y\|$ is called —.

Triangle in-equality.

(18) $\|x+y\| \leq \|x\| + \|y\|$ is called ———.

Triangle in-equality.

(19) The pair $(V, \langle \cdot, \cdot \rangle)$ is called: —

Inner Product

(20) The inner product space V , the function is given by — defined as — in V .

$$\|x\| = \sqrt{\langle x, x \rangle}, \quad x \in V, \text{ norm.}$$

(21) The two vectors are called orthogonal when they are —

$$\|x\| = \sqrt{\langle x, x \rangle}, \quad x \in V, \text{ norm.}$$

(2) The two vectors are called orthogonal when they each other ———.
Perpendicular.

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A linear function 'f' with domain $D(f)$ in a normed space is ——— iff 'f' is ———

Continuous, Bounded.

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Let $T: D(T) \rightarrow Y$ be a ——— operator where $D(T) \subseteq X$ and X, Y are normed space.

Linear.

25

Let X be a finite dimensional vector space

$D(T) \subseteq X$ and X, Y are normed space.

Linear.

(25) Let X be a finite dimensional vector space
if $x_0 \in X$ has the property $f(x_0) = 0, \forall f \in X^*$, then.

$$x_0 = 0$$